HTTP Practice Problems - Solutions

Question 1:
Objects = 5, RTT = 4 ms, Transmission time = 25 ms, Persistent HTTP

Solution:
Step 1: TCP Connection = 1 RTT = 4 ms
Step 2: First Request/Response = 1 RTT = 4 ms
Total RTT = 2 × 4 = 8 ms

Step 3: Transmission Time = 5 × 25 = 125 ms

Total Time = 8 + 125 = 133 ms

Question 2:
HTML + 2 objects = 3 objects, RTT = 3 ms, Transmission = 8 ms, Non-persistent HTTP

Solution:
Each object time = 2RTT + Transmission
= 2(3) + 8 = 14 ms

Total Time = 3 × 14 = 42 ms

Total Time = 42 ms